

DETAILED DESCRIPTION OF THE INVENTION — DATA PROCESSING, ABSTRACTED INDICES, AND OUTPUT GENERATION

Physiologic signals acquired by the intraluminal physiologic monitoring device may be processed using local, external, or distributed computational systems configured to transform raw sensor data into meaningful, non-explicit physiologic representations.

In various embodiments, signal processing may include amplification, filtering, noise reduction, normalization, temporal alignment, feature extraction, statistical modeling, and multi-sensor fusion. Processing may occur within onboard electronics, companion mobile devices, cloud-based computing environments, or hybrid processing architectures.

Sensor outputs may be converted into abstracted physiologic indices intended to represent trends or states rather than explicit anatomical information. Such indices may include hydration proxy indices, contact stability scores, compliance indicators, thermal context metrics, variability signatures, recovery trends, and other derived physiologic parameters.

Abstracted outputs may be presented visually using graphs, bars, curves, symbolic indicators, or other non-anatomical representations designed to preserve privacy while conveying physiologic insight. Visualization methods may avoid explicit anatomical depiction and instead emphasize longitudinal trends and comparative analysis.

Machine learning or adaptive modeling systems may be employed to identify patterns across time, correlate signals between sensing modalities, or generate predictive insights based on historical physiologic behavior. Models may be personalized to individual users through adaptive calibration processes or population-level comparative analysis.

Data processing systems may further support anomaly detection, trend classification, state transition detection, and contextual interpretation of physiologic signals. Outputs may be used for wellness monitoring, research applications, compatibility modeling, adaptive device configuration, or other analytic purposes.

In certain embodiments, processed data may be stored locally, synchronized with secure remote databases, or shared selectively under user-controlled permission frameworks. Encryption, anonymization, and privacy-preserving computational methods may be implemented to protect sensitive physiologic information.

The processing approaches described herein are illustrative and non-limiting. Alternative algorithms, artificial intelligence systems, computational architectures, and visualization methods capable of generating substantially similar functional outcomes are considered within the scope of the disclosure.